

Week 8 Task

Introduction to Wireless Communication

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Summary of all code:

The code implements a custom Gymnasium environment (WirelessEnv) simulating a wireless downlink where a base station serves a mobile user with the assistance of a Reconfigurable Intelligent Surface (RIS). The environment models a direct base-station-to-user link and a reflected base-station-to-RIS-to-user link, each subject to log-distance path loss and fading, and combines them to compute an overall channel gain and resulting SNR.

At each timestep, the agent selects a joint action from a discrete action space of 42 combinations (7 transmit power levels $\times$ 3 modulation schemes $\times$ RIS on/off), and receives a reward based on whether the resulting bit error rate meets a target threshold, balanced against a power-usage penalty—encouraging the agent to achieve reliable communication while minimizing transmit power.

A Deep Q-Network (DQN) agent is implemented from scratch, consisting of a replay buffer, a two-hidden-layer Q-network, and a separate target network updated periodically for training stability, with an epsilon-greedy policy that anneals exploration over training. The agent is trained for 2000 episodes and evaluated against a hand-designed greedy heuristic baseline that selects power, modulation, and RIS state based on simple SNR thresholds. Debugging during development corrected two key parameters—the RIS array size/path-loss model (so RIS reflection provides a realistic benefit) and the target SNR (set to an achievable level given the power budget)—after which the trained agent learned to consistently activate RIS and select power/modulation combinations that matched or outperformed the baseline heuristic.